Week 7: Flask: routes and templates

A minimal app

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route("/")
def index():
    return "<h1>Hello, Flask!</h1>"
```

```
flask --app app run --debug    # run in the folder
                               # that holds app.py
# open the forwarded port 5000 (Ports tab)
# stop the server: Ctrl+C
```

Browser asks → Python function answers. `--debug` reloads on every save and shows errors. Never use it on a public site.

Routes and variables

```
@app.route("/hello/<name>")
def hello(name): ...

@app.route("/square/<int:n>")   # 404 if not int
def square(n): ...

@app.route("/about")
def about(): ...
```

Converter	Matches
`<name>` (or `<string:name>`)	Any text without `/`
`<int:n>`	Whole numbers (digits only)
`<float:x>`	Decimals, **needs** a point: `2.0`
`<path:p>`	Text including `/`

Every function name must be **unique**. Missing page → 404. `abort(404)` makes one on purpose.

Folders Flask expects

```
project/
  app.py
  templates/    base.html, index.html ...
  static/       style.css, images ...
```

Templates (Jinja)

```
# in app.py
return render_template("students.html",
                       students=STUDENTS, total=5)
```

Marker	Meaning
`{{ value }}`	Print a value (escaped automatically)
`{% ... %}`	A statement: `if`, `for`, `extends`, `block`
`{# ... #}`	Comment

```
{% for s in students %}
  <li>{{ s.name }} ({{ s['class'] }})</li>
{% else %}
  <li>Nobody here.</li>
{% endfor %}

{% if score >= 50 %}Pass{% else %}Fail{% endif %}
{{ 'good' if avg >= 70 else '' }}
```

- `s.name` and `s['name']` both read a dictionary key. Use `s['class']`.
- Filters: `{{ name|title }} |upper |length |round(1) |join(', ') |default('n/a')`.
- Jinja turns `<` into `<`, so user text cannot run as HTML (stops XSS).

One layout for all pages

```
<!-- templates/base.html -->
<title>{% block title %}Site{% endblock %}</title>
<link rel="stylesheet"
  href="{{ url_for('static',
  filename='style.css') }}">
<nav><a href="{{ url_for('index') }}">Home</a></nav>
{% block content %}{% endblock %}

<!-- templates/page.html -->
{% extends "base.html" %}
{% block content %}<h1>Page</h1>{% endblock %}
```

Links and errors

```
url_for('student', student_id=3)   # /students/3
url_for('students', **{'class': '3A'})
                                   # /students?class=3A

@app.errorhandler(404)
def not_found(error):
    return render_template("404.html"), 404
```

Use `url_for('function_name')`, not typed paths. If the route changes, every link follows. Always add `, 404` to the error page, or the browser is told the page was found.

Top mistakes

- Running `flask run` from the wrong folder ("Could not import app").
- Folder not called exactly `templates` (`TemplateNotFound`).
- Forgetting `return`.
- Missing `{% endfor %}` or `{% endif %}`.